City Hospital: Data Preprocessing & Exploratory Insights

From Raw Consultation Logs (10,008 Visits) to a Clean 500-Patient Analytic Base Table (ABT)

- **Prepared by:** City Hospital Data Preprocessing (DP) Team
- **Scope:** January – June 2024 Consultations (Reference Date: July 1, 2024)
- **Deliverables:** Clean ABT (500 rows x 37 features), Data Dictionary, EDA & Roadmap

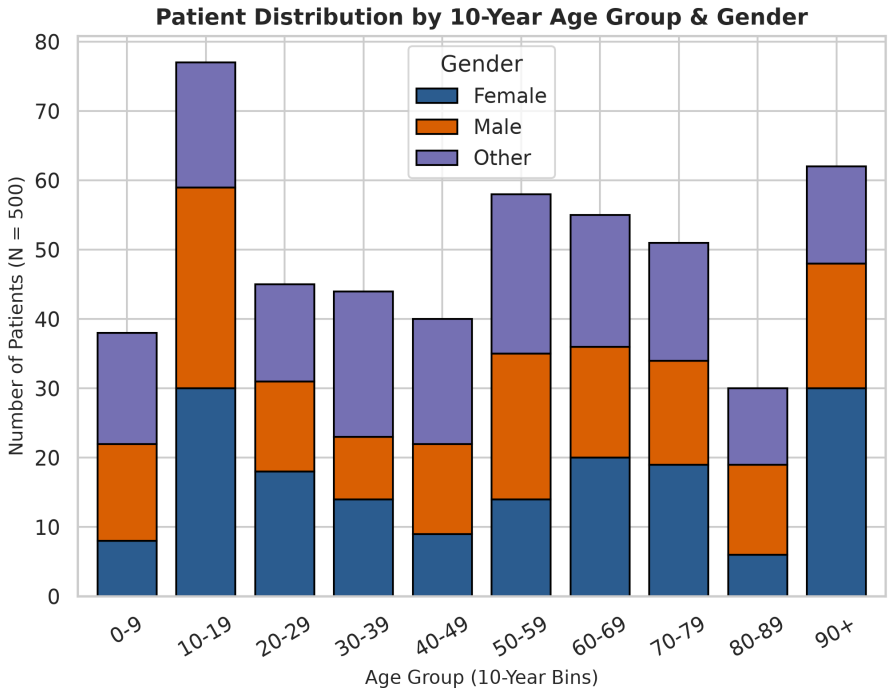
Part A: Data Quality Audit & Statistical Cleaning

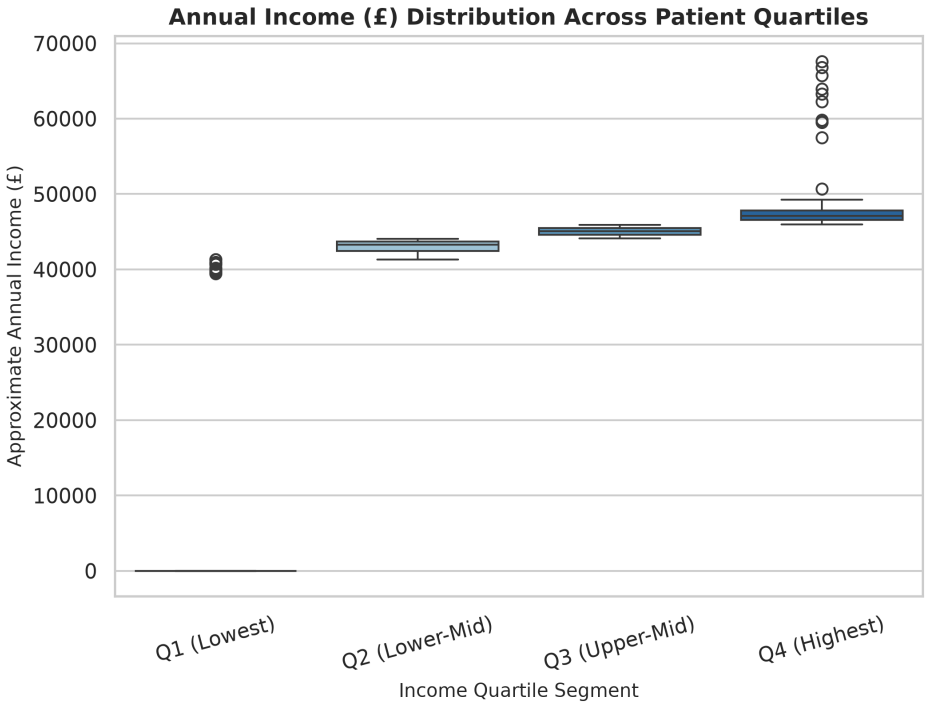
- **Zero Patient Loss:** Preserved all **500 unique patients** while removing 8 exact duplicate rows (10,000 clean visits).
- **Temporal Standardization:** Resolved conflicting static attributes (Gender, Marital Status, Profession, Age) by selecting each patient's **most recent Visit Date** record.
- **Domain & IQR Outlier Treatment:** Corrected 103 invalid Age entries (>100 yrs), capped 719 rating='6' values to 5, and Winsorized Consultation Duration (10 rows) & Price (7 rows, including £25k–£40k typos) via 1.5x IQR fences.
- **Predictive Imputation (KNNImputer k=5):** Imputed missing Age, Education, Income, and Insurance Coverage using distance-weighted KNN after assigning 'None' (Family History) and 'Uninsured' (£0 coverage policies).

Part B: Patient-Level ABT Architecture (500 Rows x 37 Features)

- **Static Profile (12 features):** Age, 10-Yr Age Group, Gender, City, Profession, Education, Marital Status, Insurance Provider, Family History, Cleaned Annual Income & Quartile.
- **Visit Behaviour (9 features):** Total Visits (mean 20.0), Distinct Departments (mean 9.9), Most Frequent Department, Avg/Total Duration, Weekend vs. Weekday Ratio (0.41).
- **Financial Exposure (7 features):** Total/Avg Consultation Price, Total/Avg Insurance Coverage, Total/Avg Out-of-Pocket Spend, and Out-of-Pocket Ratio.
- **Satisfaction & Recency (9 features):** Avg/Min/Max Rating, Low Rating Count & Share (<=2), Days Since First/Last Visit (vs. July 1, 2024), Active Span, and Visits per Month (3.33/mo).

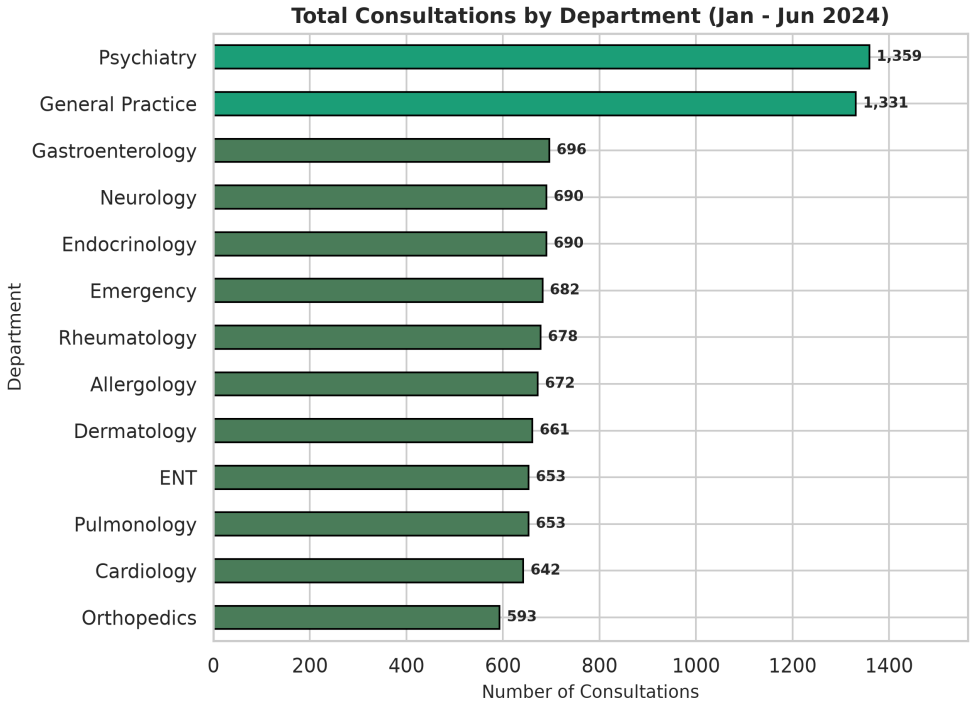
Demographics: 10-Year Age Groups & Income Quartiles

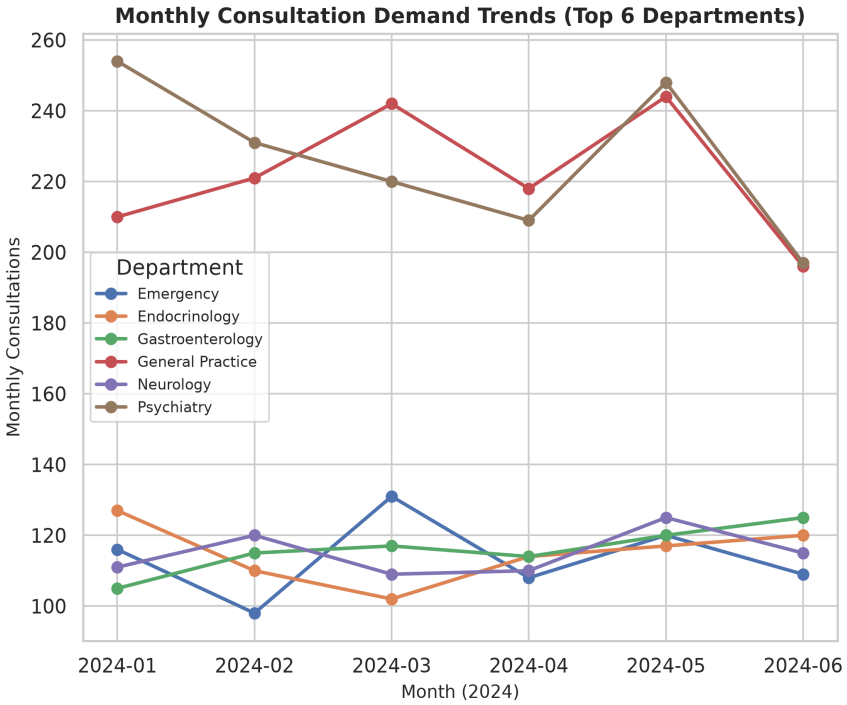




- Balanced representation across all 10-year age brackets (mean age: 49.1 yrs) and genders.
- Income quartiles cleanly separate Q1 (£0 Student/Pediatric dependents) from working/retired adults (£30k–£60k).

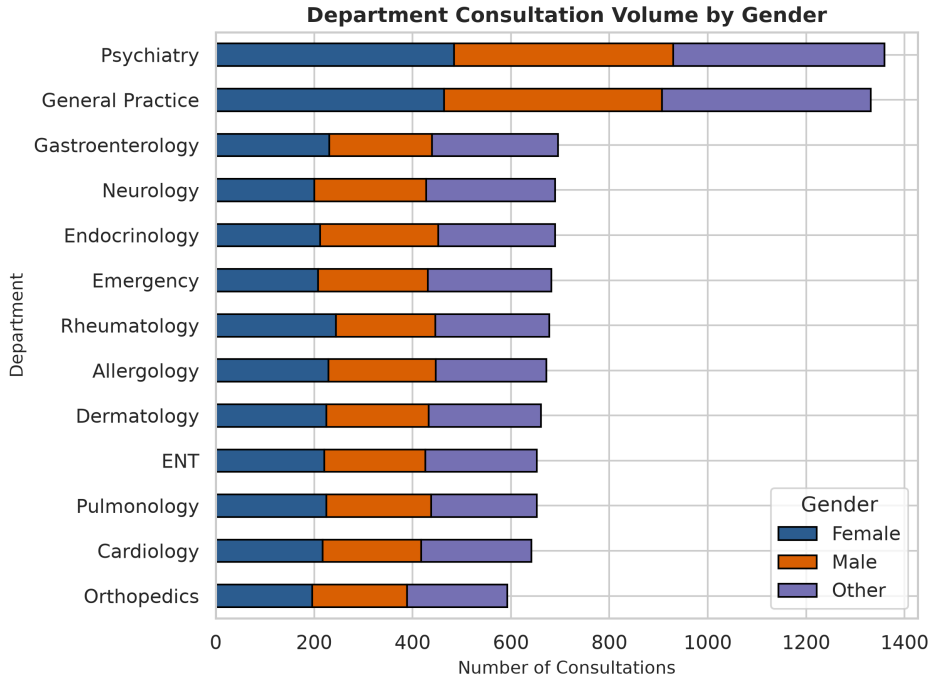
Department Demand & Monthly Trends (Jan – Jun 2024)

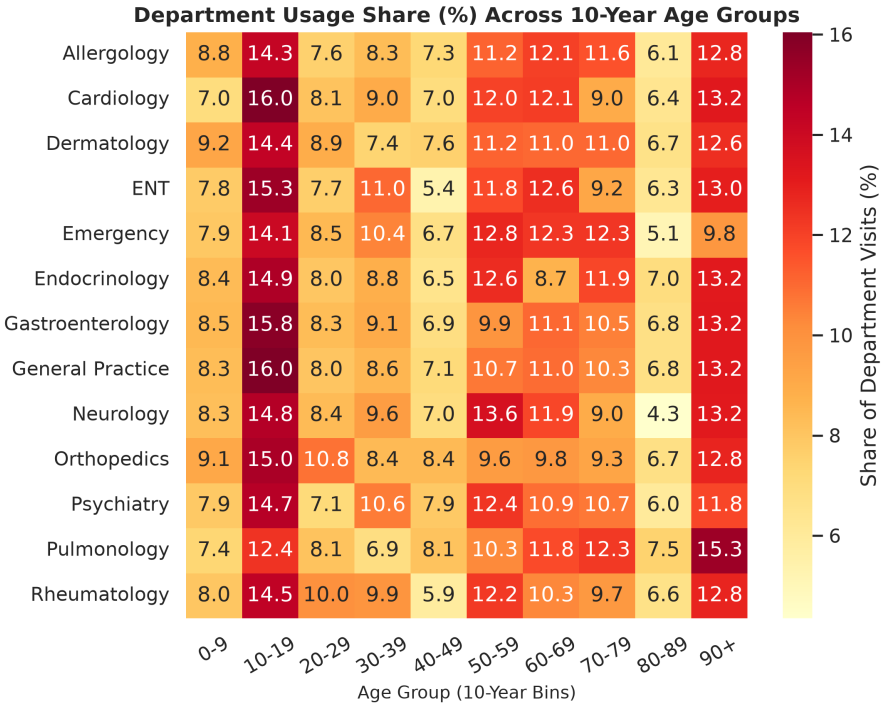




- **Psychiatry (1,359 visits)** and **General Practice (1,331 visits)** drive 26.9% of total hospital volume.
- Specialist departments average ~650 visits each; monthly demand is stable across the 6-month window.

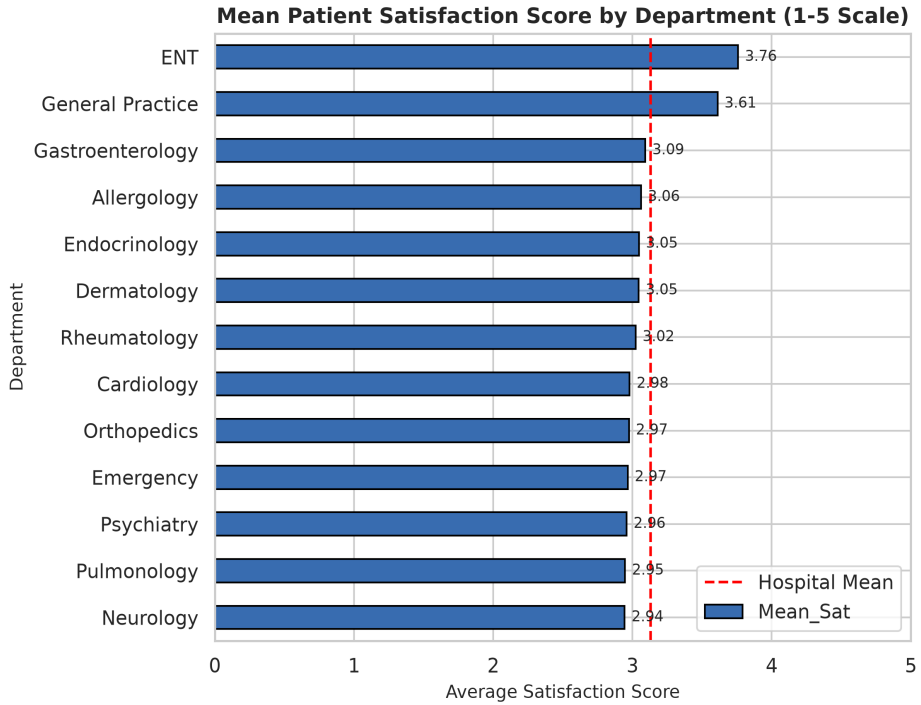
Department Utilization by Gender & Age Group

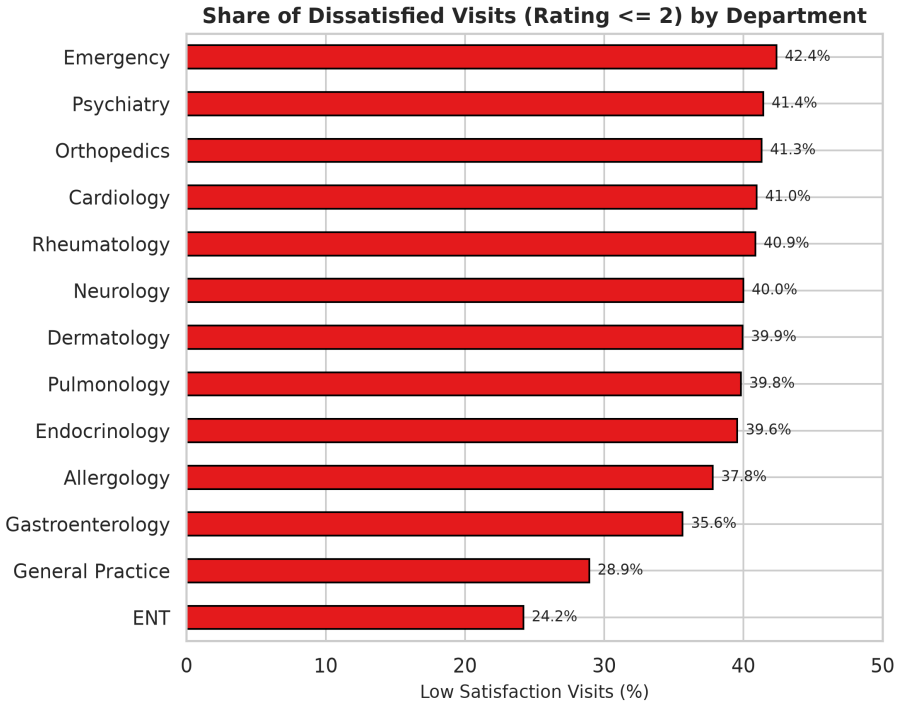




- All 13 departments see balanced utilization across Female, Male, and Other patient segments.
- High cross-department mobility: patients visit an average of 9.9 distinct departments.

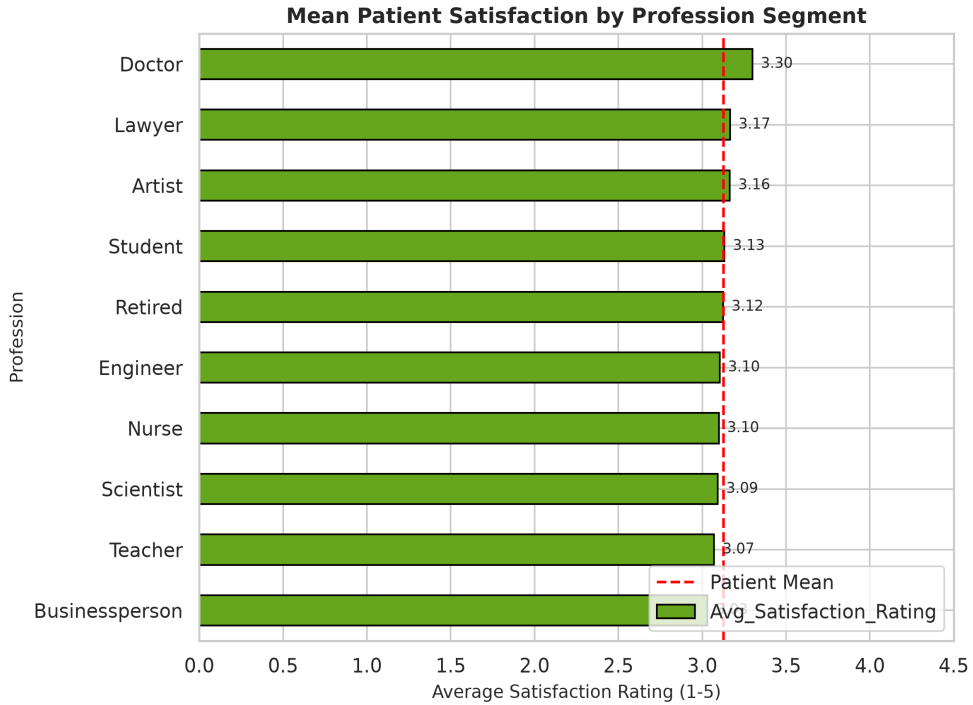
Satisfaction Pain Points Across Departments

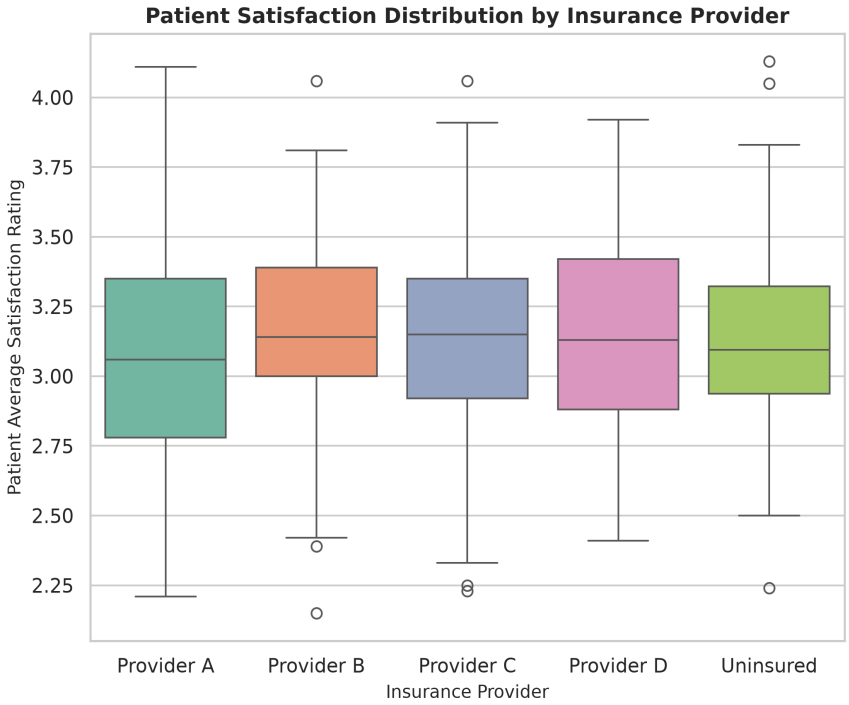




- Overall mean rating is 3.13/5.00, with 37.6% of visits rated <= 2.
- **Neurology (2.94)** has the lowest mean score; high-volume Psychiatry & GP drive the most dissatisfied visits.

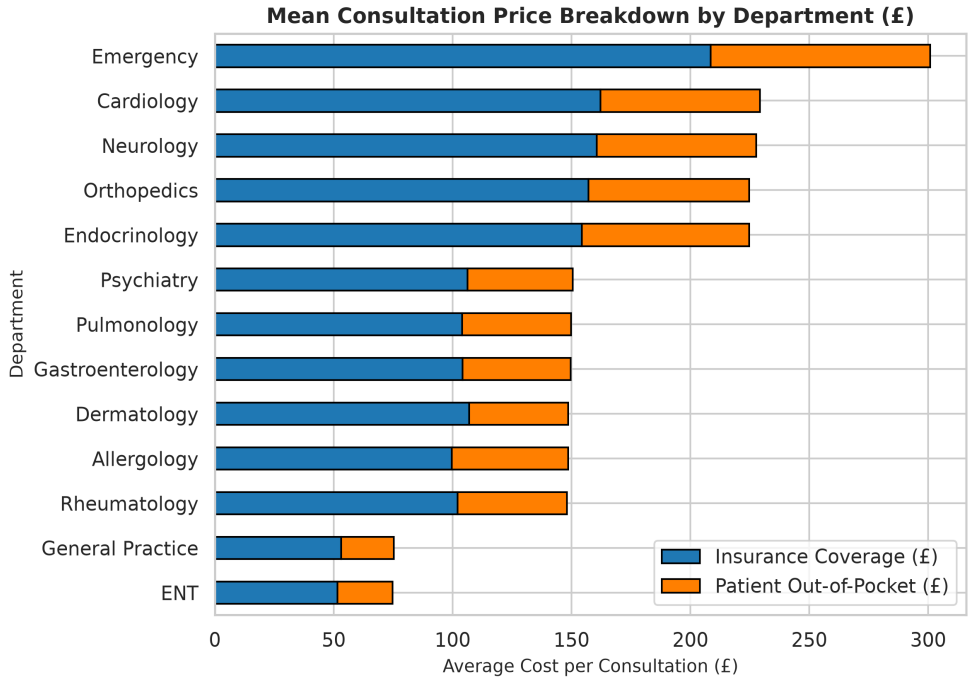
Satisfaction by Patient Occupation & Insurance Tier

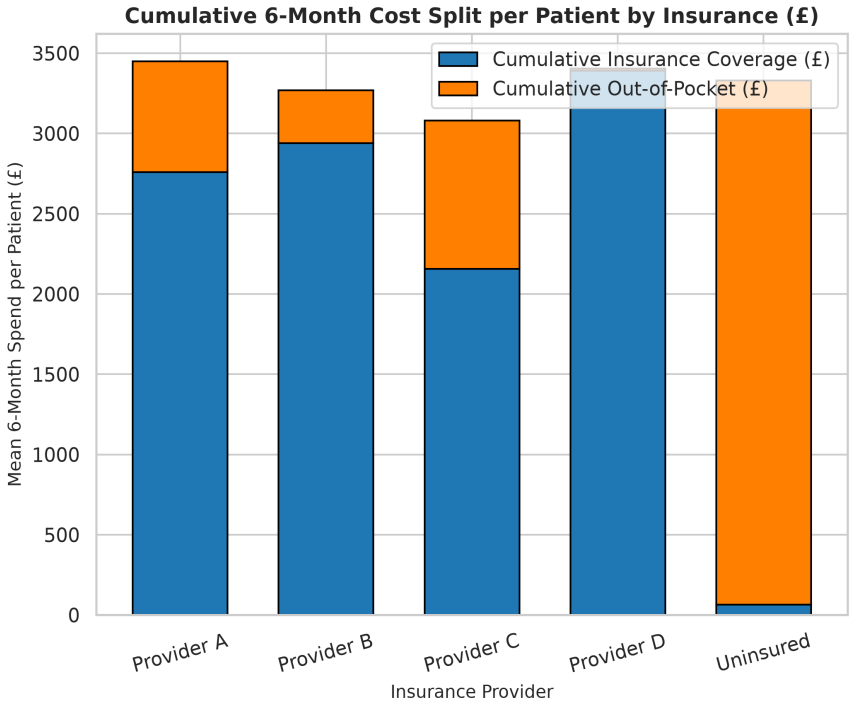




- Patient-level average satisfaction clusters around 3.15–3.25 across professions and insurance providers.
- Service quality pain points are encounter-driven rather than restricted to a single demographic class.

Financial Split: Insurance Coverage vs. Out-of-Pocket





- Insurance covers **69.8% (£1,152,924)** of gross costs; patients pay **30.2% (£499,001)** out-of-pocket.
- Emergency (£300/visit) is the costliest department; 92 Uninsured patients bear ~£3,100 in 6-month out-of-pocket costs.

Strategic Roadmap for the Next Analytics Team

- **1. Patient Segmentation (Clustering):** Segment the 500 patients by (a) *Financial Vulnerability* (Uninsured & high Out-of-Pocket Ratio), (b) *Multi-Specialty Heavy Utilizers*, and (c) *Weekend-Preferring Patients*.
- **2. Predictive Modeling Targets:** Train classification/regression models on the ABT to predict *High Dissatisfaction Risk (Share_Low_Ratings > 0.40)* and *Future Monthly Utilization (Visits_Per_Month)*.
- **3. Operational & EHR Interventions:** Expand staffing in Psychiatry and General Practice, audit service workflows in low-rated departments, and implement hard EHR input validation bounds.